

FOUNDATIONAL ERROR-CONTROL CODING RESEARCH AUDIT

FIBER-GRAND

Collapse Propositions and Kill-Fast Protocol

A preregistered scientific investment gate for certified aggregate decoding of non-invertible transformation channels

Purpose. Determine, before a multi-year investment, whether aggregate inverse-channel search has a credible route to a field-defining coding/decoding programme. The gate separates mathematical correctness, finite-block computational mechanism, code modularity, bounded-edit transfer, novelty, strong-baseline survival, and real-system relevance. A hard failure generates an explicit STOP command.

Programme under audit:	FIBER-GRAND proposal, scientific-audit revision 4
Primary starting channel:	Exactly one deletion plus independent substitutions
Computational scope:	One deletion, one insertion, two deletions, four code families
Decision classes:	CONTINUE, REVISED PIVOT, NARROW PIVOT, STOP
Release:	v1.0, August 2026

Abstract

The proposal under audit makes a correct and important observation: on a non-invertible random-transformation channel, maximum-likelihood (ML) decoding ranks candidate inputs by the *sum* of all latent-history probabilities that generate the observation, not by the probability of the first or best individual history. The one-deletion-plus-substitution likelihood and its simple unseen-candidate bound are mathematically sound. The proposal’s field-defining case, however, cannot rest merely on defining a transformation fiber, restating posterior candidate rank as “fiber guesswork,” or applying a generic branch-and-bound certificate. Conditional guesswork with side information is classical, and admissible best-first/priority-first search, alignment summation, weighted transducers, and synchronization decoding have substantial prior art.

This dossier identifies the narrower scientific residue that could still be foundational: a code-modular aggregate inverse-search engine whose *fully accounted* exact work is substantially below codeword-wise ML; a theory of search inflation and ambiguity overhead; and a tractability boundary for bounded-edit or bounded-drift channels. It proves the exact membership-first form of the one-deletion decoder, generalizes the unseen bound to fixed edit-position stream families, and specifies a reproducible experiment suite with preregistered GO/NARROW/STOP thresholds. The computational package checks exactness exhaustively, measures how often pathwise first-hit is wrong, compares history-driven and prefix-driven certified search against exhaustive codeword ML, tests one insertion and two deletions, and transfers the same channel engine across random-linear, CRC-like, Hamming, and VT codebooks. A positive standard run authorizes only a revised claim contract; it cannot by itself establish novelty, superiority over specialized synchronization decoders, or field-defining status.

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1 Proposal reconstruction and scientific boundary

1.1 The proposal's central claim

Let a latent transformation history $Z = z$ act on an input x through T_z , producing $Y = T_z(x)$. For fixed (x, y) , define the inverse history fiber

$$\mathcal{F}_y(x) = \{z : T_z(x) = y\}, \quad (1)$$

and its aggregate score

$$\Lambda_y(x) = \sum_{z \in \mathcal{F}_y(x)} p(z | x) = W(y | x). \quad (2)$$

For a uniform prior over a codebook $\mathcal{C}$, exact ML decoding is

$$\hat{c}_{\text{ML}}(y) \in \arg \max_{c \in \mathcal{C}} \Lambda_y(c). \quad (3)$$

The proposal correctly distinguishes this from ranking candidates by

$$M_y(x) = \max_{z \in \mathcal{F}_y(x)} p(z | x), \quad (4)$$

which is a best-path rather than best-candidate criterion.

1.2 What is scientifically correct and immediately useful

The following statements survive direct audit.

- C1. Multiplicity can reverse the candidate order.** For $\mathcal{C} = \{000, 001\}$, observation 00, and one uniform deletion with no substitutions, $\Lambda_{00}(000) = 1$ while $\Lambda_{00}(001) = 1/3$, although every compatible individual deletion history has mass $1/3$.
- C2. The fixed-observation order condition is exact.** A pathwise first-hit rule is ML for every codebook if and only if its induced candidate order never places a lower- Λ_y candidate before a higher- Λ_y candidate.
- C3. The general lazy certificate is valid.** If an incumbent codeword lower bound exceeds every discovered-competitor upper bound and a valid upper bound on every undiscovered candidate, the incumbent is exact ML.
- C4. The one-deletion likelihood is linear-time per candidate.** All deletion-alignment mismatch counts can be updated by a one-step recurrence.
- C5. The simple multi-stream unseen bound is valid.** If η_j is the largest unprocessed component in deletion stream j , every completely unseen candidate has score at most $\sum_j \eta_j$.

1.3 What cannot carry a field-defining novelty claim

Three conceptual corrections are mandatory.

Candidate rank is conditional guesswork. Under a uniform prior on X , Bayes' rule gives

$$P_{X|Y}(x | y) \propto W(y | x) = \Lambda_y(x). \quad (5)$$

Therefore the proposal's

$$G_{\text{fib}}(X | Y) = \text{rank}_{\Lambda_Y}(X) \quad (6)$$

is exactly the optimal posterior guessing rank of X given side information Y . Guessing with side information and its conditional Rényi characterization were studied by Arıkan and subsequently sharpened nonasymptotically by Sason and Verdú [3, 4]. The term “fiber guesswork” may remain a channel-specific interpretation, but not a new generic random variable.

The stopping lemma is a branch-and-bound certificate. The inequality comparing an incumbent lower bound to frontier or residual upper bounds is mathematically useful, but it is structurally a standard admissible-search certificate. Sequential synchronization decoding, priority-first ML decoding, weighted automata, and best-first search all provide relevant prior foundations [5, 6, 8]. Novelty must lie in a new bound, a new complexity theorem, or a new code-modular implementation advantage—not in the abstract certificate alone.

Summing hidden alignments is established inference semantics. Forward summation over deletion/insertion alignments, pair-HMM paths, and weighted-transducer paths is established. ML deletion analyses explicitly depend on embedding/alignment multiplicity [7, 9, 10]. The proposal's defensible distinction is not “sum the paths,” but rather:

aggregate inverse-channel search + generic code interface + runtime exact certificate + work-tail theory.

(7)

2 Minimal mathematical development

2.1 One deletion plus substitutions

Let $x = (x_1, \dots, x_n) \in \{0, 1\}^n$. Each bit is independently flipped with probability $p < 1/2$, after which exactly one location J is deleted with probability q_j . The observation has length $n - 1$. For deletion position j , define

$$d_j(x, y) = \sum_{i < j} \mathbf{1}\{x_i \neq y_i\} + \sum_{i > j} \mathbf{1}\{x_i \neq y_{i-1}\}. \quad (8)$$

Then

$$\Lambda_y(x) = \sum_{j=1}^n q_j p^{d_j(x, y)} (1 - p)^{n-1-d_j(x, y)}. \quad (9)$$

All d_j are obtained in $O(n)$ time from

$$d_1(x, y) = \sum_{i=2}^n \mathbf{1}\{x_i \neq y_{i-1}\}, \quad (10)$$

$$d_{j+1}(x, y) = d_j(x, y) - \mathbf{1}\{x_{j+1} \neq y_j\} + \mathbf{1}\{x_j \neq y_j\}. \quad (11)$$

2.2 History streams and the unseen-candidate bound

For each j , enumerate surviving-bit substitution patterns $e \in \{0, 1\}^{n-1}$ in nonincreasing BSC mass, and both possible deleted input bits $b \in \{0, 1\}$. The inverse candidate is

$$x = \text{Ins}_j(y \oplus e, b), \quad (12)$$

with component mass

$$\omega_j(e) = q_j p^{w_H(e)} (1 - p)^{n-1-w_H(e)}. \quad (13)$$

Let $\eta_j(t)$ be the next unprocessed mass in stream j .

Theorem 1 (Unseen-candidate bound and exact stopping). *At any search time t , every candidate never emitted by any stream satisfies*

$$\Lambda_y(x) \leq U_t^{\text{new}} := \sum_{j=1}^n \eta_j(t). \quad (14)$$

Assume every discovered complete candidate has been membership-tested and every discovered codeword has been exactly scored. Let

$$s_t^* = \max_{c \in \mathcal{C} \cap D_t} \Lambda_y(c), \quad T_t^* = \{c \in \mathcal{C} \cap D_t : \Lambda_y(c) = s_t^*\}, \quad (15)$$

where D_t is the set of discovered candidates. If $\mathcal{C} \cap D_t \neq \emptyset$ and

$$s_t^* > U_t^{\text{new}}, \quad (16)$$

then T_t^ is exactly the global ML tie set. In particular, the ML codeword is unique if and only if $|T_t^*| = 1$.*

Proof. For an unseen x , its unique component in each alignment stream has not been emitted and therefore has mass at most that stream's next mass $\eta_j(t)$. Summing the n alignment contributions proves the bound. Every discovered codeword has its exact score represented in s_t^* and T_t^* , while discovered noncodewords are outside the optimization domain. The strict inequality excludes every undiscovered codeword from tying or exceeding s_t^* , so the global ML tie set is precisely T_t^* . $\square$

2.3 Membership-first refinement

The proposal's literal Algorithm 1 exactly scores every newly discovered candidate before checking code membership. That work is unnecessary.

Proposition 1 (Membership-first exactness). *Assume (i) every newly discovered complete candidate is subjected to an exact code-membership test, (ii) every discovered codeword is exactly scored, and (iii) U_t^{new} remains a valid bound on every completely undiscovered candidate. Then discovered noncodewords need not be exactly scored. Stopping when the best discovered-codeword score exceeds U_t^{new} remains exact ML.*

Proof. A discovered noncodeword is not in the maximization domain $\mathcal{C}$. The only possible competitors are discovered codewords, which are exactly scored, and undiscovered codewords, each bounded by U_t^{new} . $\square$

This refinement is essential for the code-agnostic claim: with a cheap membership oracle, complete likelihood evaluation is performed only for the small subset of emitted candidates that are codewords.

2.4 Prefix aggregate search

For a candidate prefix u , maintain alignment-specific mismatch counts accumulated over processed input symbols. Because $p < 1/2$, every unprocessed surviving symbol contributes at most $1 - p$. Thus

$$B(u) = \sum_{j=1}^n q_j p^{m_j(u)} (1 - p)^{n-1-m_j(u)} \quad (17)$$

is an admissible upper bound on every full candidate extending u , where $m_j(u)$ counts observed mismatches already forced under deletion alignment j . Best-first expansion with an optional prefix-feasibility oracle is exact when the incumbent score exceeds the largest frontier bound.

The computational package distinguishes:

- **PREFIX-L0**: no code-specific prefix pruning;
- **PREFIX-L2**: systematic-linear prefix feasibility, representing code-assisted acceleration rather than membership-only universality.

2.5 Fixed edit-position generalization

For exactly t uniform deletions, there are $\binom{n}{t}$ deletion-position streams. Each stream emits a surviving substitution pattern and t hidden input bits. The same proof yields

$$U_t^{\text{new}} = \sum_{S: |S|=t} \eta_S. \quad (18)$$

The naive number of streams is $\Theta(n^t)$ for fixed t . This is polynomial for each fixed t but is an XP dependence, not a proof of fixed-parameter tractability $f(t)n^c$. Any FPT claim requires a different merged-state or combinatorial argument.

One insertion has an analogous stream family indexed by insertion position, with the observed inserted symbol probability included in the component mass.

3 Collapse propositions

The programme should be evaluated conjunctively. A fatal failure cannot be averaged away by a favorable score elsewhere.

Gate	Necessary proposition	Required early evidence	Failure action
F0	The surviving synthesis is not already equivalent to established conditional guesswork, A*/priority-first decoding, weighted-transducer inference, or a known synchronization decoder.	Independent claim-by-claim primary-source audit.	Stop the field-defining claim or rename/narrow.
F1	The aggregate likelihood, inverse generators, membership-first rule, and certificates are exactly correct.	Exhaustive micro-world agreement; zero false certificates.	Immediate hard STOP until repaired.
F2	Path multiplicity changes decisions often enough to justify aggregation.	Pathwise first-hit disagrees with aggregate ML in multiple nondegenerate configurations.	Stop the aggregation programme; ordinary first-hit suffices in the tested domain.

Gate	Necessary proposition	Required early evidence	Failure action
F3	The first complete channel has a strong code-agnostic computational mechanism.	One deletion plus substitutions: exact completion, controlled tails, and large total-work reduction on at least two unmodified code families.	Stop the broad programme.
F4	The inverse search does not pay back its gain through hidden expansion.	Search inflation relative to ideal posterior rank and history-per-candidate overhead remain controlled.	Tighten bounds once; if unchanged, STOP or narrow.
F5	Code modularity is operational rather than rhetorical.	Same channel engine works on random-linear, CRC-like, Hamming, and nonlinear/VT membership oracles.	Reclassify as a code-specific synchronization decoder.
F6	The mechanism transfers beyond the single hand-picked channel.	Positive one-insertion and two-deletion results under the same certificate discipline.	Narrow to one deletion; do not claim a general transformation paradigm.
F7	The gain survives strong exact and specialized baselines.	Priority-first/A*, channel-code trellis where exact, VT/GC+/marker/polar synchronization methods, and codeword-side likelihood search receive equal channel knowledge.	If gains disappear, STOP the field-defining programme.
F8	A genuinely new theorem exists beyond generic posterior guesswork and branch-and-bound.	Search-inflation law, ambiguity-dependent complexity bound, FPT/X-P/hardness boundary, or certified resource converse.	Algorithm paper only; no field-defining theory claim.
F9	At least one calibrated physical impairment lies in the favorable regime.	Post-front-end timing-slip, deletion/duplication, variable-demapping, or storage trace with fair end-to-end comparison.	Theoretical/special-purpose result only.

Investment discipline. F1 is a binary correctness gate. F2–F4 decide whether the core mechanism exists. F5–F6 decide breadth. F7–F9 decide whether the work can plausibly become field-defining. The released standard package can close only F1–F6. A positive computational result therefore emits a *revised pivot*, not an unconditional field-defining continuation.

4 Computational programme

4.1 E0: exactness and implementation audit

The package checks:

1. one-deletion recurrence, dynamic program, vectorized codeword scoring, and exact rational scoring;

2. one-insertion likelihood against independent vectorized scoring;
3. complete history-driven decoding for one and two deletions;
4. prefix-driven decoding and certificate validity;
5. random-linear, CRC-like, Hamming, and VT membership/codeword construction;
6. zero false certificates in exhaustive small observation spaces.

Any discrepancy produces

```
STOP_FIELD_DEFINING_FIBER_PROGRAMME
STAGE=EXACTNESS_OR_CERTIFICATE_FAILURE
```

4.2 E1: theory and novelty-boundary audit

The code verifies three finite identities:

1. aggregate candidate order equals posterior order under a uniform prior;
2. the random-code conditional success law is

$$\mathbb{P}\{\hat{X} = X \mid G = g\} = \frac{\binom{N-g}{M-1}}{\binom{N-1}{M-1}}; \quad (19)$$

3. the minimal one-deletion multiplicity counterexample is reproduced exactly.

The first two are scientifically useful but not field-defining novelty.

4.3 E2: path-versus-fiber boundary and search inflation

For small n , enumerate the full ambient input space and compute exact aggregate likelihood ranks. For each random trial, record:

$$I_{\text{search}}^{\text{opt}} = \frac{Q_{\text{cand}}}{G_{\text{posterior}}^{\text{optimistic}}}, \quad (20)$$

$$I_{\text{search}}^{\text{pess}} = \frac{Q_{\text{cand}}}{G_{\text{posterior}}^{\text{pessimistic}}}, \quad (21)$$

$$I_{\text{hist}} = \frac{Q_{\text{hist}}}{Q_{\text{cand}}}. \quad (22)$$

The test includes uniform and edge-biased deletion positions and substitution probabilities $p \in \{0, 0.05, 0.10\}$. It also measures the frequency with which pathwise first-hit returns a non-ML codeword.

4.4 E3: one deletion plus substitutions

At $n \in \{12, 16, 20, 24\}$ and rate near 0.75, compare:

- HISTORY-L0: membership-only history enumeration with membership-first exact scoring;
- PREFIX-L0: code-independent prefix bound without prefix pruning;
- PREFIX-L2: prefix feasibility for systematic linear codes;
- exhaustive codeword-wise exact ML.

Use random-linear and CRC-like codes, $p \in \{0.02, 0.05, 0.10\}$, and per-block records sufficient to recompute all means and tails.

4.5 E4–E5: bounded-edit transfer

E4 tests one insertion plus substitutions at the same lengths and rates. E5 tests two deletions at $n \in \{10, 12, 14, 16, 18, 20\}$. These are not substitutes for a general transducer theorem; they ask whether the same aggregate-search principle survives two structurally different extensions.

4.6 E6: code-family transfer

Use one-deletion plus substitutions on:

- random systematic linear code;
- deterministic CRC-like cyclic code;
- Hamming (15, 11) code;
- nonlinear VT code with a checksum membership oracle.

The same channel history engine must remain exact. PREFIX-L2 is reported only where the code exposes the required interface.

5 Cost accounting and preregistered thresholds

5.1 Work vector

The package does not report “number of guesses” alone. It records

$$\mathbf{W} = (Q_{\text{hist}}, Q_{\text{cand}}, Q_{\text{mem}}, Q_{\text{score}}, Q_{\text{symbol}}, Q_{\text{heap}}, Q_{\text{bound}}, Q_{\text{prefix}}, M_{\text{peak}}, T_{\text{wall}}). \quad (23)$$

Three scalar work models—optimistic, balanced, and pessimistic—apply frozen weights to these components. The scientific record retains the full vector, so a reviewer can replace the scalarization.

5.2 Reference baseline

The initial computational gate uses exhaustive codeword-wise aggregate ML as an independent exact oracle and a first work reference. This is necessary for correctness and for establishing whether candidate-likelihood evaluation is avoided. It is *not* sufficient for a final field-defining claim. The next contract must include exact priority-first/code-trellis methods and specialized synchronization systems.

5.3 Core thresholds

The standard profile requires, at $p = 0.02$ and $p = 0.05$:

- one deletion: at least two code families, completion at least 0.95, mean total-work speedup at least 20, fifth-percentile speedup at least 2, median candidate-score reduction at least 100, and favorable work-slope separation;
- one insertion: at least one family with mean speedup at least 15;
- two deletions: at least one family with mean speedup at least 10;
- code transfer: at least three families passing at $p = 0.05$;
- boundary: multiple pathwise-error configurations, median search inflation no greater than 20, and controlled history inflation.

These are early-investment thresholds, not claims of universal superiority.

6 Decision logic

6.1 STOP

Emit

STOP_FIELD_DEFINING_FIBER_PROGRAMME

when exactness fails, path multiplicity is operationally irrelevant, or the one-deletion core fails to produce a strong exact computational signal across two code families.

6.2 Narrow pivot

Emit

PIVOT_TO_NARROW_ONE_EDIT_FIBER_TRACK

when the first one-deletion problem passes but insertion/two-deletion/code-transfer evidence does not. This authorizes one narrowly scoped theorem/algorithm paper, not a general random-transformation programme.

6.3 Revised pivot

Emit

PIVOT_TO_REVISSED_FIBER_TRACK

when F1–F6 pass. The original wording must still stop because “fiber guesswork” is not a new general information-theoretic object. The revised programme is:

certified code-modular aggregate inverse search + ambiguity/search-inflation theory + tractability boundary.

(24)

6.4 Full continuation

A full field-defining continuation is deliberately not machine-authorized by the present release. It additionally requires:

1. independent primary-source novelty adjudication;
2. strong exact and specialized baseline survival;
3. a new decoder-specific theorem beyond the generic certificate;
4. a calibrated real synchronization impairment and end-to-end advantage.

7 Real use cases and model discipline

The proposal’s strongest application candidates are not generic wireless links but systems in which the receiver truly observes a small, probabilistically weighted alignment uncertainty:

1. **Short-packet timing slips:** missed/duplicated symbol decisions or bounded sample-clock drift, with the synchronizer exporting an alignment lattice rather than a single hard alignment.
2. **Variable-length demapping:** unequal-length labels or framing errors create insertion/deletion uncertainty; candidate demapping histories are aggregated jointly with FEC membership.

3. **Storage and DNA:** deletion/insertion/substitution channels and short strands provide natural benchmark domains, though they do not by themselves establish telecommunications relevance.
4. **Packet-sequence transformations:** bounded loss, duplication, and reordering with outer checksum/network-code membership.

A physical claim requires measured post-front-end statistics. Merely renaming a deletion channel as “clock slip” is insufficient.

8 Scientific interpretation of a positive early result

A positive standard run would establish four things:

1. aggregation is operationally necessary because pathwise first-hit can be wrong;
2. the simple certificate can stop far before codebook-wide likelihood evaluation in favorable short, high-rate regimes;
3. membership-first scoring makes the L0 code interface genuinely useful;
4. the same mechanism transfers at least to one insertion, two deletions, and unrelated codebooks.

It would *not* establish:

- that the method beats the best specialized decoder;
- an asymptotic complexity exponent advantage;
- polynomial-time exact decoding for unrestricted transducers;
- novelty of posterior guesswork;
- a real telecommunications deployment advantage.

The correct outcome in that situation is a revised, sharply de-risked programme rather than either premature celebration or premature termination.

9 Authorized next contract after a revised pivot

Only the following four tasks are authorized after a positive computational gate.

9.1 N1: independent novelty matrix

For every proposed theorem or algorithmic claim, identify the closest primary source in conditional guesswork, sequential synchronization decoding, priority-first ML decoding, weighted transducers, ambiguous probabilistic automata, and deletion/insertion decoding. Classify each claim as distinct, partially anticipated, known, or unresolved.

9.2 N2: strong baseline gate

Implement or obtain reproducible baselines for:

- priority-first/ A^* decoding on a code trellis;
- exact channel-code forward or list methods where feasible;
- VT, Guess-and-Check/GC+, marker/watermark, and synchronization-polar methods in their matched regimes;
- align-then-decode and best-path approximations;
- codeword-wise aggregate likelihood and any exact list-pruning alternative.

The revised programme stops if its fully accounted advantage disappears.

9.3 N3: theorem extraction

Target one of the following, in order of value:

1. a nontrivial bound on $Q_{\text{cand}}/G(X | Y)$ or $Q_{\text{hist}}/Q_{\text{cand}}$ in terms of alignment ambiguity;
2. a strict fixed-parameter/XP boundary for bounded deletions or drift;
3. a hardness theorem showing where exact aggregate candidate search ceases to be favorable;
4. a certified resource-constrained converse under an explicit oracle model.

9.4 N4: calibrated use-case gate

Use one real or high-fidelity front-end model. Freeze the channel law and compare total latency, memory, reliability, and tail behavior. Do not begin FPGA/ASIC work before N1–N3 survive.

10 Reproducibility contract

The package includes:

- complete source code and tests;
- frozen profile and decision contract;
- exact micro-world audits;
- compressed per-trial tables and uncompressed summaries;
- figures and machine-readable verdict;
- SHA-256 manifests;
- a WSL launcher that creates an isolated environment, runs the gate, and publishes only the experiment package to GitHub.

An installation, dependency, authentication, or push failure must be reported as

`EXECUTION_BLOCKED_NOT_A_SCIENTIFIC_VERDICT`

and never converted into a scientific STOP.

11 Reference-standard release validation

The frozen standard profile was executed independently before release. Fourteen unit tests and all six higher-level exactness groups passed. The machine classification was

```
PIVOT_TO_REVISED_FIBER_TRACK
STAGE=COMPUTATIONAL_CORE_PASSED_NOVELTY_REFRAME_REQUIRED
```

The decisive finite-block evidence is summarized in Table 2. Operation-count speedup uses the frozen balanced work vector and exhaustive codeword-wise aggregate ML as the initial exact reference. Wall-clock measurements are Python/NumPy implementation measurements on the validation machine and are reported separately.

Table 2: Reference-standard early computational signal. Values are finite-block observations, not asymptotic or hardware claims.

Test	Frozen operation signal	Python wall-clock signal	Interpretation
Path/fiber bound-ary, $n = 10\text{--}12$	Search inflation at most 5.08 in median across frozen configurations	Not a speed benchmark	Pathwise first-hit disagreement reached 0.75; aggregation is operationally necessary rather than merely semantic.
One deletion, $p = 0.05$, $n = 24$	Mean speedup 1.31×10^4 ; completion 1.00	Median speedup about 83	Median 71 histories; 99th percentile 1014; median membership fraction 1.79×10^{-4} .
One insertion, $p = 0.05$, $n = 24$	Mean speedup 1.78×10^5 ; completion 1.00	Median speedup about 6.3×10^2	Median 109 histories; 99th percentile about 1393.
Two deletions, $p = 0.05$, $n = 20$	Mean speedup 3.09×10^2 ; completion 1.00	Median speedup about 1.87	Positive but modest wall-clock gain at the largest tested length; only two largest- n trials were used in this kill-fast probe.
Code transfer, $p = 0.05$, $n = 15\text{--}16$	Mean speedups about $1.75 \times 10^2\text{--}5.16 \times 10^2$	HISTORY-L0 mean speedups about 1.59–2.59	Random-linear, CRC-like, Hamming, and VT membership interfaces all remained exact.

The evidence is unusually strong for the one-deletion and one-insertion computational mechanism. It is not yet a field-defining result for four reasons. First, the exact reference is exhaustive codeword-wise likelihood evaluation rather than the full set of strongest specialized baselines. Second, the two-deletion wall-clock gain is modest and based on a deliberately small largest-length sample. Third, the generic information-theoretic interpretation of “fiber guesswork” is already conditional posterior guesswork. Fourth, no calibrated physical synchronization trace has been tested.

Accordingly, the original claim set must stop, while a revised and sharply bounded research path is scientifically justified. The next investment gate is N1–N4, not a deep rerun of the same synthetic experiments.

12 Conclusion

The attached proposal is scientifically stronger than a superficial “GRAND for deletions” description. It identifies the right ML semantic object and supplies a valid exact certificate for a controlled non-invertible

channel. Its broad field-defining language nevertheless overstates the novelty of candidate rank and generic bounded search. The kill-fast protocol therefore tests the computational core aggressively while preregistering a mandatory theoretical reframe.

The programme deserves further investment only if exact aggregate search simultaneously demonstrates path-multiplicity relevance, controlled search inflation, strong work reduction, code transfer, and bounded-edit transfer. Even then, the next stage is not hardware or a general transducer library. It is a narrow novelty-and-baseline confrontation followed by extraction of a genuinely new complexity theorem. If those tests fail, the programme must stop immediately.

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